

ClearShift

No one performs regulated work without proven qualification.

Qualification gating, pre-shift safety, and compliant crew planning on Databricks.

Talking points: ClearShift is a working build on the Databricks Platform for manufacturers with hazardous, regulated work. It answers one question the learning system cannot: is tomorrow's shift compliantly staffed, and if not, what do we do about it before the shift starts.

ClearShift closes manufacturing's largest uncontrolled safety exposure, using data you already have, for \$275K to \$440K a year plus protection against a \$1M-plus event.

Your systems record who was trained. They cannot tell you who is cleared to work tomorrow, so unqualified work reaches the floor and audit readiness is rebuilt by hand. ClearShift closes that gap three ways:

- **Gate it before the shift.** Every scheduled seat is checked against what it requires, so exposure is caught the evening before, not after an incident.
- **Prove it on demand.** Audit readiness is derived from dated records, cutting 40 hours per plant to minutes.
- **Plan it intelligently.** A model flags who will lapse and an optimizer staffs a compliant crew from the workforce you already have.

Talking points: This is the whole story on one slide. Situation: only qualified people may do hazardous work. Complication: your systems record training, not tomorrow's clearance. Answer: ClearShift, worth this range annually with catastrophic-risk protection on top. Everything after this slide is evidence for these three points: prevent, prove, plan.

Your systems record who was trained, not who is cleared to work tomorrow.

- A worker can be scheduled tomorrow into work they are not qualified to perform.
- The learning system records course completion, not who is cleared to work a seat.
- Audit readiness is reconstructed by hand, plant by plant, the week an auditor calls.
- First-time hazardous work slips through without a supervisor present.
- The gap is invisible until an incident or a citation makes it visible.

Talking points: Every manufacturer with confined space, hot work, or lockout/tagout carries this exposure daily. The data exists, in HR, the scheduler, the LMS, but it is fragmented, so no one can answer the operational question in time to act.

One uncontrolled exposure runs \$165K in penalties and over \$1M in a fatality event.

\$165K

per willful or repeat OSHA citation (2024 maximum). Serious violations run to \$16,550 each.

>\$1M

all-in for a single confined-space fatality citation and the litigation behind it.

40 hrs

per plant, per audit, spent reconciling rosters by hand today.

Talking points: The average-case value is real: penalties avoided plus audit labor. But the tail risk is what funds this. One prevented confined-space incident dwarfs the cost of the platform. This is a control, not only an efficiency play.

ClearShift turns fragmented qualification data into a governed daily control.

FOUNDATION — All qualification, schedule, and requirement data, unified and effective-dated on Databricks.

GOVERNANCE — Site and supervisor are the access model. Every answer is scoped and traceable to its source.

INTELLIGENCE — A model predicts who will lapse; an optimizer plans a compliant crew; Genie answers in plain language.

Talking points: Same arc as any Databricks data intelligence story: get the foundation right, govern it, then make it intelligent. The difference here is the intelligence makes a compliance decision, not a dashboard.

Every seat is checked the evening before, so exposure never reaches the floor.

- Tomorrow's schedule, checked against what each job and work center requires.
- One verdict per person: cleared, first performance, or not cleared, with a plain reason.
- Requirements are effective-dated, so the check reflects what was required on the day.
- Catches the case a clock-in gate misses: a worker moved mid-shift into work they are not qualified for.

Talking points: This is the daily driver. A supervisor sees tomorrow's crew before the shift, not after. The reason is in plain language so it is actionable, not a red cell on a spreadsheet.

Audit readiness is proven on demand, cutting 40 hours per plant to minutes.

- Every regulated requirement at a site, and whether it is currently held.
- Card serials, trainers, evidence references, ready to spot-check.
- The answer is derived from dated records, so historical clearance is provable.

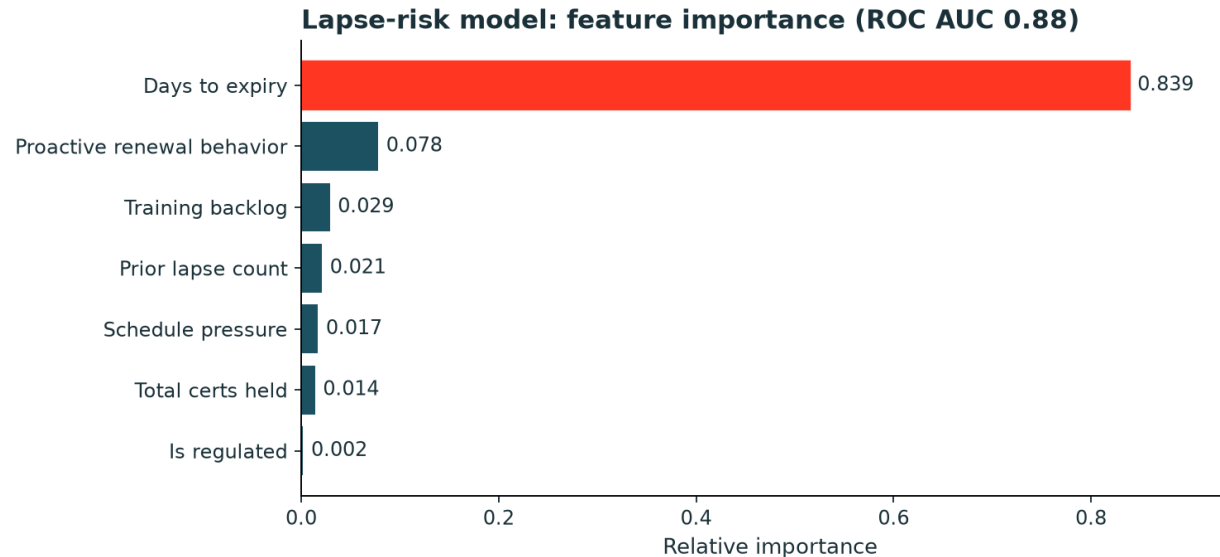
From 40 hrs to minutes

audit roster reconciliation, per plant.

Talking points: When the auditor arrives, the roster is already the answer. No fire drill. This is where the audit-labor savings land, and where a finding becomes a non-event.

A model flags who will lapse before the shift, not after they expire.

- A gradient-boosted model scores every worker–certification pair: probability the certification lapses before the next hazardous job.
- Leading signals: days to expiry, renewal behavior, training backlog, prior lapses, schedule pressure.
- Measured performance: ROC AUC 0.88. Expiry proximity carries most of the signal; behavior adds the early warning.



Talking points: A rule tells you who is already expired. The model tells you who is about to lapse and why, in time to renew before the shift is exposed. This is real machine learning, not an expiry date lookup.

The optimizer staffs a compliant crew from your workforce, and names what it cannot.

- A CP–SAT optimizer assigns the qualified pool to tomorrow's regulated seats.
- Hard compliance rules are constraints, never traded off: hold every required qualification, supervise every first performance, no double–booking.
- Among compliant plans, it prefers workers whose own certification is least likely to lapse.
- Seats it cannot staff are named, with the reason, so you act before the shift.

What-if: build tomorrow's compliant crew The lapse-risk model scores every worker; a CP-SAT optimizer assigns the qualified pool to tomorrow's regulated seats, never breaking a compliance rule, preferring workers whose own certification is least likely to lapse. Seats it cannot compliantly staff are named, with the reason, so you act before the shift.		
4 Seats compliantly staffed	2 Exposed seats now cleared	3 Cannot staff compliantly
Solve tomorrow's crew → OPTIMAL - total assigned lapse risk 0.12. Hard compliance rules are constraints, never traded off.		
The plan		
re-planned	Danielle Okafor BURN DEPT - Confined Space Entry + Hot Work / Welding + Lockout / Tagout + Respiratory Protection was Marcus Ellery — Not cleared	lapse risk 0.04
re-planned	Marcus Ellery PLASMA TABLE - Fall Protection + Hot Work / Welding was Trent Boland — Not cleared	lapse risk 0.03
on schedule	Rafael Ibarra INDIRECT - TOOLING - FIXTURING - Crane and Hoist Signalling + Lockout / Tagout	lapse risk 0.02
on schedule	Yvette Coleridge STOCKROOM- SHIPPING- RECEIVING - Powered Industrial Truck	lapse risk 0.03
Still exposed — act before the shift		
Not cleared	BAY OPERATIONS - MATERIALS Powered Industrial Truck Qualified capacity is fully committed to other regulated seats this shift.	
First performance	BURN DEPT Confined Space Entry + Hot Work / Welding + Lockout / Tagout + Respiratory Protection First-time regulated work needs a qualified supervisor present; none available at Peoria IL - Plant 3081.	
Not cleared	SPIN DEPT Abrasives Blast and Coating + Respiratory Protection No worker holds the full requirement (Abrasives Blast and Coating + Respiratory Protection) for SPIN DEPT at Peoria IL - Plant 3081.	

Talking points: This is the decision, not a report. In the live scenario it clears exposed seats by moving the right qualified people, and it tells you exactly which seats have no compliant answer and why. Scarce low-risk capacity gets spent where it matters.

Anyone can ask in plain language and see the exact query behind the answer.

- Natural-language questions over the governed gold tables.
- Returns the SQL it ran, so the answer is inspectable, not taken on trust.
- Answers within the caller's scope, and declines questions outside the data.

The screenshot shows the ClearShift assistant interface. At the top, it says "ClearShift assistant" and "Grounded in work center requirements, credential state and the forward schedule". A user asks: "Ask about tomorrow's exposure, site audit readiness, thin cover, or who is one credential away from closing a gap. Answers are limited to your scope." The assistant responds: "How many workers have a certification expiring in the next 90 days, by site?" The assistant then provides a detailed answer: "By site, **Peoria IL - Plant 3081** has **1** worker with a certification expiring in the next 90 days. In the visible results, **Peoria IL - Plant 3081** is also the highest site count at **1** worker." Below this, it shows the SQL query: "SELECT 'location', COUNT(DISTINCT 'employee_id') AS 'workers_expiring_next_90_days' FROM 'horizontal_dev_serverless_catalog', 'clearshift', 'gold_renewal_pipeline' WHERE 'days_until_expiry' IS NOT NULL AND 'days_until_expiry' BETWEEN 0 AND 90 AND 'location' IS NOT NULL GROUP BY 'location' ORDER BY 'workers_expiring_next_90_days' DESC, 'location'". A table shows the results: LOCATION | WORKERS_EXPIRING_NEXT_90_DAYS. Below the table, it says "Answer with Governed queries" and lists several questions: "Who is blocked from working tomorrow?", "How many workers have a certification expiring in the next 90 days, by site?", "Is Peoria ready for the audit?", "Where is cover thin?", "Who could cover blast and coating?", "What lapses in the next 90 days?", "Who is overdue on compliance training?", and "Which manager has the worst completion?". At the bottom, it says "Grounded in" and lists the tables: gate.v_shift_clearance, gate.v_site_roster, gate.v_coverage, gate.v_credential_opportunity, and gate.v_renewal_pipeline.

Talking points: For the questions no one anticipated. A safety lead asks in plain English and gets an answer grounded in the same governed tables, with the query shown. Governance does not stop at the dashboard.

ClearShift moves incident rate, citations, backfill spend, and time-to-staff.

RECORDABLE INCIDENT RATE — Fewer unqualified-work exposures reaching the floor.

CITATIONS AND FINDINGS — Audit readiness proven on demand, not reconstructed.

BACKFILL SPEND — Compliant substitutes found from the existing workforce, not overtime or contractors.

TIME TO STAFF A COMPLIANT SHIFT — From a manual scramble to a solved plan.

~\$275K to \$440K

estimated annual value for a twelve-plant manufacturer, before tail risk.

Talking points: For the executive sponsor, the number and the tail risk. For the domain owner, the daily workflow: clearance, audit readiness, and the what-if plan. Both get what they came for.

One dataset flows through every layer, on one platform.

Lakeflow ingest → **Unity Catalog** govern → **Lakebase** serve → **ML** predict → **CP-SAT** decide → **Genie** query → **Databricks App** surface.

No siloed parts. Raw operational data becomes a compliance decision and a plain-language answer.



One dataset flows through every layer · no siloed parts

Talking points: The integration is the point. Raw operational data becomes a compliance decision and a plain-language answer, on one platform, with governance that travels the whole way.

Start where exposure is highest, prove it in weeks, then expand.

- Start with the foundation you already have: HR, scheduler, LMS.
- Prove audit readiness in weeks, not a rebuild.
- Add the model and the optimizer where the exposure is highest.
- A reference implementation, adapted to your plants, not deployed as-is.

Talking points: Close on the path. This is not a rip-and-replace. It sits beside the systems you run and answers the question they cannot. Start where the exposure is, prove it, expand.